

Part 1 – Data Analytics Insights Report: Smart City Traffic Intelligence

The analysis examined 48,204 hourly traffic and weather observations from the westbound I-94 corridor near Minneapolis–St Paul. SQLite, statistical analysis and Power BI were used to identify traffic patterns, congestion behaviour and relationships with weather. Overall, traffic demand varies substantially by time, while temperature alone has only a weak relationship with traffic volume.

Key Findings

Average traffic volume was 3,259.82 vehicles per hour, with a median of 3,380 and standard deviation of 1,986.84, indicating considerable variation in traffic demand. Traffic ranged from 0 to 7,280 vehicles per hour.

Annual recorded traffic volume fluctuated considerably and reached its highest total in 2017 at 35.43 million vehicles. However, annual comparisons should be treated cautiously because the number of observations differs substantially between years. Some changes therefore reflect unequal data coverage rather than changes in traffic demand alone.

The 2017 hourly analysis showed a clear time-of-day pattern. Traffic was lowest overnight, increased sharply during the morning period and reached its strongest average level at approximately 4 PM. This suggests that time of day is an important factor for traffic-management planning.

Temperature showed only a weak positive correlation with traffic volume ($r = 0.1303$). Higher traffic occurred across a broad range of temperatures, suggesting that temperature alone isn't a strong predictor of traffic demand.

Weather conditions showed some association with traffic. Clouds had the highest average traffic volume at 3,618.45 vehicles, while Squall had the lowest at 2,061.75, a difference of 1,556.70 vehicles. Congestion, defined as traffic volume above 5,500 vehicles, occurred in 14.73% of observations. The odds of congestion were approximately 26.5% lower during clear weather than cloudy weather, although this association doesn't establish causation.

Data Quality and Implications

Power Query identified no null values or type errors. However, the temperature analysis revealed readings around -273.15°C (0 Kelvin), which are physically implausible and likely represent sensor errors. These values should be treated during the data-cleaning stage before predictive modelling.

For the smart-city mobility team, the results suggest that time-based traffic patterns should be a primary consideration for congestion management. Monitoring and commuter information could be concentrated around morning and afternoon peak periods. Weather information may provide additional context but should be combined with hour, day, holiday and seasonal factors rather than used alone.

Conclusion

Part 1 shows that traffic demand on the I-94 corridor is highly variable and strongly influenced by temporal patterns, while weather provides additional but more limited information. The next stage should clean the identified sensor anomalies and combine time, weather and calendar features to create a reliable foundation for the Python pipeline and subsequent machine-learning models.